FYP Progress Update: Test Results and Proposed Next Steps

1 October 2026

A learned market-making policy (a gradient-boosted model) made **+3.55% net of fees** on the held-out test months (Aug–Nov 2025), with a 95% confidence interval of +2.51% to +4.60% over 34,086 simulated fills. Adding the STGAT's forecasts to it did not improve it. I would like your view on the proposed next step before continuing.

1 Headline result

The learned policy passed its pre-registered verdict (pessimistic-fill return above 0, with the event-clustered 95% CI excluding 0). Every strategy below was run once on the test months with its settings frozen in advance.

Strategy (test, Aug–Nov 2025)	Return, pessimistic	95% CI	Net P&L	Return, strict	Fills / events
G: learned policy (GBM)	**+3.55%**	+2.51 to +4.60%	**+$3,748**	+0.39%	34,086 / 1,325
G+S: learned policy + STGAT features	+2.90%	+1.88 to +3.85%	+$3,235	+0.01%	35,759 / 1,313
Quote every eligible order (no model)	−0.34%	−0.77 to +0.10%	−$1,133	−3.07%	110,867 / 1,609
A10: hand-set maker rules	+1.80%	−48 to +62%	+$1.87	+28.5%	83 / 29

- **The model does the work.** The same set of quotes posted without the model loses money; the model's selection of which quotes to post makes +$3,748.
- **It held up out of sample.** Validation (Jun–Jul 2025) gave +3.73%; test gave +3.55%. The earlier rule-based maker (A10) fell from +21.7% to +1.8% over the same step.
- **It trades at scale.** 34,086 fills, against 83 for the rule-based maker, so the result does not hinge on a few events.

2 How it was tested

The policy predicts, for each candidate maker quote, its net P&L per contract at settlement after fees, and posts the quote only if the prediction exceeds 1¢.

1. **Data and split:** Kalshi MECE markets, Nov 2024 – Nov 2025, split 7 / 2 / 4 months (train Nov–May, validation Jun–Jul, test Aug–Nov), as you approved.
2. **Model:** scikit-learn `HistGradientBoostingRegressor` (histogram gradient boosting, LightGBM-style), trained on 163,128 simulated fills from the training months only. Inputs are 14 features known at the quoting hour: order side and price, spread, mid, recent trades, 24-hour price change, basket context (basket size, sum of mids, best other leg), hour, fee status and category.
3. **Simulator:** a tape-driven replay of real Kalshi trades and hourly quotes, with inventory limits and Kalshi's fee schedule. Fills are bounded by three queue-position models: optimistic, **pessimistic (the pre-registered verdict)** and strict (competing makers also improve their price).

4. **Pre-registration:** the model, features, threshold, validation gate and verdict were committed before any test-month run; the frozen model is hash-checked before scoring.
5. **Leak checks:** with shuffled training labels the edge disappears (+0.05¢ per contract); it survives removing time-to-close, fee status and category, and excluding sports markets (whose recorded close time is the actual match end).
6. **Validation gate inside the simulator:** +3.73% (CI +2.43% to +5.03%) on Jun–Jul 2025, before the single test run.

The test months were used earlier for the rule-based maker; this is disclosed, and no design choice for the learned policy used them.

3 Where the STGAT adds value

The STGAT forecasts basket prices well, but its forecasts carry almost no information for trading beyond simple market features.

Use of the STGAT	Result (pre-registered test unless noted)	Why
Forecasting basket prices	**Beats persistence (0.635× MSE) and the shrinkage oracle (0.922×)**	Anticipates convergence to $1 as baskets settle
Next-hour signal for the rule-based maker (S4/S5)	No improvement; the filter never fired	The maker's P&L is decided at settlement; even perfect next-hour foresight adds about $1
Extra inputs to the learned policy (G+S)	**−0.65 pp vs G** (CI −1.40 to +0.09)	Signal inflated in-sample: correlation with later price moves 0.062 on training months, 0.016–0.025 out of sample
Forecasting settlement outcomes (exploratory, development data)	No model beats the market price (log-loss 0.4589 vs 0.4577)	Kalshi prices are already well calibrated to outcomes

This answers the thesis question directly: the learned model adds value in forecasting basket prices, but the trading profit comes from execution (which quotes to post given spread, favourite bias and informed flow), which a model on simple market features captures.

4 Other routes tested

Every route that relied on knowing more than the market failed; the only lasting edges were in execution.

Route	Outcome
Kalshi → Polymarket lead (pre-registered, fresh 2026 data, 768 matched soccer events)	Kalshi leads Polymarket: **+0.70¢ per contract, CI +0.47 to +0.97**, but Polymarket's sports taker fees turn it into −0.53¢ to −1.35¢
Weather brackets vs live station observations	Kalshi reprices impossible brackets within the hour (99.2% bid below 2¢ at the first quote)
Predicting which basket quotes fill completely	Fills are predictable (AUC 0.91), but the predictable fills lose money (adverse selection)
Spotify #1-song markets vs streaming data	Not testable: no free daily streaming history for 2024–25
Rule-based maker A10 on the test months	+1.8%, not significant; its opportunities collapsed from about 63 to 5–7 eligible leg-days per month by Oct–Nov 2025

The literature agrees: frontier AI models lost money trading Kalshi in 2026 (Prediction Arena), and Kalshi's weather prices beat public forecasts (Crosier, 2026).

5 Limitations

The main caveat is competition: under the strict fill model the edge is not significant.

- **Strict fills:** if competing makers also improve their price, the learned policy's return falls to +0.39% (CI −0.96% to +1.78%). The claim is "profitable under pessimistic fills", with strict reported as the bound.
- **Backtest, not live trading:** orders were never placed. Fills are inferred from historical trades; market impact, faster competitors and larger order sizes are not modelled.
- **Hourly resolution:** the simulator re-quotes once an hour; a live maker reacting in seconds could be picked off more often.
- **Reused test months:** Aug–Nov 2025 had already been used for the rule-based maker. Disclosed; no learned-policy choice used them.
- **Short window:** 13 months in total, two of them validation; one regime of a fast-growing market.

6 Proposed next step: a model benchmark

I propose a small, pre-registered benchmark: do state-of-the-art models beat the gradient-boosted policy on the same task? Estimated 2–4 days, leaving about two and a half weeks for writing.

Challenger	Why include it	Expectation
CatBoost	Strong gradient boosting with native categorical handling	Close to the current model
A modern deep tabular model (TabM or FT-Transformer)	The strongest neural challengers to boosting on tabular data	Likely below boosting (Grinsztajn et al., NeurIPS 2022)
STGAT with a policy head	Tests whether the graph model itself can be the trading policy	Likely below boosting, given the results above

Each model is tuned on validation only, then all are run once on the test months and reported, whatever they show. Deep reinforcement-learning market makers are left out: too data-hungry and unstable for the time left.

7 Questions for you

1. Does a learned (gradient-boosted) policy that is profitable out of sample meet the "positive P&L with a model" requirement, or must the STGAT itself be the trading model?
2. Is it acceptable to report profitability under the pessimistic fill model, with the strict-competition result (not significant) stated as a limitation?
3. Should I run the model benchmark above, or go straight to writing up with the current results?